

Здание 17 - по ядру Linux (кросскомпиляция)

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1) Устанавливаем компилятор

```
svikari@svikari-VirtualBox:~$ sudo apt update
sudo apt install gcc-arm-linux-gnueabi
[sudo] password for svikari:
Hit:1 https://dl.google.com/linux/chrome-stable/deb stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
```

2) Делаем конфиг на ARM и собираем

```
svikari@svikari-VirtualBox:~/linux$ ARCH=arm make CROSS_COMPILE=arm-linux-gnueabi multi_v7_defconfig
#
# configuration written to .config
#
```

```
svikari@svikari-VirtualBox:~/linux$ ARCH=arm CROSS_COMPILE=arm-linux-gnueabi make zImage -j$(nproc)
SYNC      include/config/auto.conf.cmd
SYSHDR    arch/arm/include/generated/uapi/asm/unistd-oabi.h
SYSHDR    arch/arm/include/generated/uapi/asm/unistd-eabi.h
WRAP      arch/arm/include/generated/uapi/asm/kvm_para.h
WRAP      arch/arm/include/generated/uapi/asm/bpf_perf_event.h
WRAP      arch/arm/include/generated/uapi/asm/errno.h
WRAP      arch/arm/include/generated/uapi/asm/ioctl.h
WRAP      arch/arm/include/generated/uapi/asm/ipcbuf.h
WRAP      arch/arm/include/generated/uapi/asm/msgbuf.h
WRAP      arch/arm/include/generated/uapi/asm/param.h
WRAP      arch/arm/include/generated/uapi/asm/poll.h
WRAP      arch/arm/include/generated/uapi/asm/resource.h
```

```
AS       arch/arm/boot/compressed/dsinit.o
AS       arch/arm/boot/compressed/bswapsdi2.o
AS       arch/arm/boot/compressed/piggy.o
LD       arch/arm/boot/compressed/vmlinux
OBJCOPY  arch/arm/boot/zImage
Kernel: arch/arm/boot/zImage is ready
svikari@svikari-VirtualBox:~/linux$ ls arch/arm/boot
bootp      deflate_xip_data.sh  Image      Makefile
compressed dts                  install.sh  zImage
svikari@svikari-VirtualBox:~/linux$
```

```
svikari@svikari-VirtualBox:~/linux$ ARCH=arm make dtbs -j$(nproc)
SYNC    include/config/auto.conf.cmd
DTC     arch/arm/boot/dts/airoha/en7523-evb.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-cubieboard6.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-guitar-bb-rev-b.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-labrador-base-m.dtb
DTC     arch/arm/boot/dts/allwinner/sun4i-a10-a1000.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-roseapplepi.dtb
DTC     arch/arm/boot/dts/allwinner/sun4i-a10-ba10-tvbox.dtb
DTC     arch/arm/boot/dts/actions/owl-s500-sparky.dtb
DTC     arch/arm/boot/dts/allwinner/sun4i-a10-chuwi-v7-cw0825.dtb
DTC     arch/arm/boot/dts/allwinner/sun4i-a10-cubieboard.dtb
```

```
svikari@svikari-VirtualBox:~/linux$ ls arch/arm/boot/dts/arm/vexpress*.dtb
arch/arm/boot/dts/arm/vexpress-v2p-ca15_a7.dtb
arch/arm/boot/dts/arm/vexpress-v2p-ca15-tc1.dtb
arch/arm/boot/dts/arm/vexpress-v2p-ca5s.dtb
arch/arm/boot/dts/arm/vexpress-v2p-ca9.dtb
```

3) Копируем в свою папку и запускаем через QEMU

```
svikari@svikari-VirtualBox:~/linux$ ls -l ~/labs/
total 11772
-rw-rw-r-- 1 svikari svikari 14329 jun 17 21:13 vexpress-v2p-ca9.dtb
-rwxrwxr-x 1 svikari svikari 12034560 jun 17 21:11 zImage
```